

Web Server using Docker

Objective

Deploy and manage an Nginx web server using Docker and understand container lifecycle and management.

Software Required

Docker Desktop, WSL2, Command Prompt.

Docker commands used:

- `docker pull nginx`
- `docker images`
- `docker run -d -p 8080:80 --name mywebserver nginx`
- `docker ps`
- `docker logs mywebserver`
- `docker exec -it mywebserver bash`
- `ls`
- `exit`
- `docker stop mywebserver`
- `docker start mywebserver`
- `docker stop mywebserver`
- `docker rm mywebserver`
- `docker rmi nginx`

Command Explanation

The project downloads the Nginx image, creates a container, publishes port 8080, verifies the running container, checks logs, opens a shell inside the container, then stops, starts, removes the container and finally deletes the image.

1. Download the Nginx Image

```
C:\Users\mrecw>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
1645c1e06f46: Pull complete
df68ee7e7a00: Pull complete
cd9307c9ecd8: Pull complete
1b30016634d5: Pull complete
fcb6fd84b2a0: Pull complete
acf093e7a04f: Pull complete
e95a6c7ea7d4: Pull complete
1cf7d051b485: Download complete
e2c07e54e55a: Download complete
Digest: sha256:ec4ed8b5299e5e90694af7750eb6dff2627317d30544d056b0371f8082f7bce
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview nginx
```

2. Check Downloaded Images

```
C:\Users\mrecw>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	 Info →	 In Use
nginx:latest	ec4ed8b5299e	241MB	66MB	EXTRA	

3. Create the Web Server Container

```
C:\Users\mrecw>docker run -d -p 8080:80 --name mywebserver nginx
25d3d8df8d486e208ce22b3a99a3fe28497b142380cec48efa0574403cdf17d1
```

4. Check Running Container

```
C:\Users\mrecw>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
25d3d8df8d48	nginx	"/docker-entrypoint..."	53 seconds ago	Up 53 sec
ports 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp		mywebserver		

5. Open the Web Server

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

6. View Logs

```
C:\Users\mrecw>docker logs mywebserver
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/01 17:15:49 [notice] 1#1: using the "epoll" event method
2026/07/01 17:15:49 [notice] 1#1: nginx/1.31.2
2026/07/01 17:15:49 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/01 17:15:49 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
2026/07/01 17:15:49 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/01 17:15:49 [notice] 1#1: start worker processes
2026/07/01 17:15:49 [notice] 1#1: start worker process 29
2026/07/01 17:15:49 [notice] 1#1: start worker process 30
2026/07/01 17:15:49 [notice] 1#1: start worker process 31
2026/07/01 17:15:49 [notice] 1#1: start worker process 32
2026/07/01 17:15:49 [notice] 1#1: start worker process 33
2026/07/01 17:15:49 [notice] 1#1: start worker process 34
2026/07/01 17:15:49 [notice] 1#1: start worker process 35
2026/07/01 17:15:49 [notice] 1#1: start worker process 36
2026/07/01 17:15:49 [notice] 1#1: start worker process 37
2026/07/01 17:15:49 [notice] 1#1: start worker process 38
2026/07/01 17:15:49 [notice] 1#1: start worker process 39
2026/07/01 17:15:49 [notice] 1#1: start worker process 40
2026/07/01 17:15:49 [notice] 1#1: start worker process 41
2026/07/01 17:15:49 [notice] 1#1: start worker process 42
2026/07/01 17:15:49 [notice] 1#1: start worker process 43
2026/07/01 17:15:49 [notice] 1#1: start worker process 44
172.17.0.1 - - [01/Jul/2026:17:17:15 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36 Edg/149.0.0.0" "-"
2026/07/01 17:17:15 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://localhost:8080/"
172.17.0.1 - - [01/Jul/2026:17:17:15 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36 Edg/149.0.0.0" "-"
```

7. Enter the Container

```
C:\Users\mrecw>docker exec -it mywebserver bash
root@25d3d8df8d48:/# is
bash: is: command not found
root@25d3d8df8d48:/# ls
bin    docker-entrypoint.d  home  media  proc  sbin  tmp
boot  docker-entrypoint.sh lib   mnt    root  srv   usr
dev    etc                  lib64 opt    run   sys   var
root@25d3d8df8d48:/# exit
exit

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug mywebserver
  Learn more at https://docs.docker.com/go/debug-cli/
```

8. Stop the Container

```
C:\Users\mrecw>docker stop mywebserver
mywebserver
```

9. Start the Container Again

```
C:\Users\mrecw>docker start mywebserver
mywebserver
```

10. Remove the Container

```
C:\Users\mrecw>docker stop mywebserver
mywebserver

C:\Users\mrecw>docker rm mywebserver
mywebserver
```

11. Remove the Image

```
C:\Users\mrecw>docker rmi nginx
Untagged: nginx:latest
Deleted: sha256:ec4ed8b5299e5e90694af7750eb6dffd2627317d30544d056b0371f8082f7bce
```

Observations

Docker successfully isolated the web server inside a container. Port mapping allowed access through `http://localhost:8080`. Logs and shell access confirmed that the container was functioning correctly.

Conclusion

This experiment demonstrated Docker fundamentals including image management, container deployment, networking, monitoring, and lifecycle operations. It provided practical experience with containerized web server deployment.